

TESLA

Factors Influencing Tesla Resale Prices

A Business Analytics Study

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Software Used
Microsoft Excel and Python (Google Colab)

1. Executive Summary

This study examines the factors associated with the resale prices of used Tesla vehicles in the United States. The analysis uses a secondary dataset containing 1,579 vehicle-level observations and 16 variables, obtained through Kaggle and identified by its publisher as data collected from publicly available Tesla.com information. The study focuses on vehicle model, manufacturing year and mileage as the main explanatory variables, with sold price as the dependent variable.

Descriptive analysis shows substantial variation in resale prices. The average sold price is \$68,391.45, while the median is \$60,400. Model-wise differences are pronounced: Model 3 has the lowest average price at \$53,875.50, while Model S has the highest at \$97,636.60. Correlation analysis indicates a moderate positive association between vehicle year and price ($r = 0.380$) and a moderate negative association between mileage and price ($r = -0.345$).

Simple regression confirms that both year and mileage are statistically significant predictors when examined separately. The multiple regression model, incorporating model, year and mileage simultaneously, is statistically significant and explains approximately 89.37% of the variation in sold price ($R^2 = 0.893708$). One-way ANOVA and the Kruskal–Wallis non-parametric test both provide strong evidence of statistically significant price differences across Tesla model groups.

The findings indicate that model, year and mileage are important observable factors associated with Tesla resale prices.

2. Problem Statement

The resale price of a used Tesla vehicle can vary considerably even within the same broad product family. Buyers need to assess whether a quoted price is consistent with observable vehicle characteristics, while sellers, dealers and pricing analysts need a systematic basis for setting or evaluating resale prices. Without an analytical framework, pricing decisions may rely heavily on subjective judgement or simple comparisons with listings.

This study therefore addresses the following business problem: understanding which observable vehicle characteristics are significantly associated with Tesla resale prices. The analysis concentrates on three variables available consistently in the dataset—vehicle model, manufacturing year and mileage—and evaluates their individual and combined relationships with sold price.

3. Research Question

“What factors significantly influence the resale price of Tesla vehicles?”

4. Objectives

- To describe the distribution and variation of Tesla resale prices using descriptive statistics and graphical analysis.
- To examine the relationships between Tesla resale price, manufacturing year and mileage using correlation and simple regression analysis.
- To determine whether vehicle model, manufacturing year and mileage significantly explain variation in resale price using multiple regression analysis.
- To test whether resale prices differ significantly across Tesla models using one-way ANOVA.
- To confirm model-wise differences using the Kruskal–Wallis non-parametric test and integrate the evidence into business implications.

5. Hypotheses

All statistical tests use a significance level of $\alpha = 0.05$.

Hypothesis	Null hypothesis (H ₀)	Alternative hypothesis (H ₁)	Test
H ₁	Vehicle model, manufacturing year and mileage do not significantly explain variation in resale price.	At least one selected characteristic significantly explains variation in resale price.	Multiple regression
H ₂	Mileage has no significant relationship with resale price.	Mileage has a significant relationship with resale price.	Simple regression
H ₃	Vehicle year has no significant relationship with resale price.	Vehicle year has a significant relationship with resale price.	Simple regression
H ₄	Mean resale prices are equal across Tesla models.	At least one Tesla model has a different mean resale price.	One-way ANOVA
H ₅	Resale-price distributions are the same across Tesla models.	At least one Tesla model has a different resale-price distribution.	Kruskal–Wallis

6. Data and Methodology

6.1 Data Source

The study uses the dataset Daily Used Tesla Car Sales for United States, obtained through Kaggle and published by Saturn Data Cloud. The dataset contains vehicle-level information such as VIN, year, model, colour, mileage, trim, sold price, interior, wheels, features, location and sold date. The publisher identifies the underlying source as Tesla.com. The dataset is treated as secondary, web-sourced data; Tesla is not claimed as the direct provider of the compiled Kaggle dataset.

URL : [kaggle.com/datasets/saturndatacloud/daily-used-tesla-car-sales-for-united-states](https://www.kaggle.com/datasets/saturndatacloud/daily-used-tesla-car-sales-for-united-states)

6.2 Sample and Variables

The dataset contains 1,579 vehicle-level observations and 16 variables. Sold price is the dependent variable, while model, manufacturing year and mileage are the primary explanatory variables. Model 3 is the reference category in the multiple regression.

Variable	Role	Description
sold_price	Dependent variable	Final sold price of the used Tesla vehicle, USD
model	Categorical predictor	Tesla model: m3, ms, mx, my
year	Numeric predictor	Vehicle manufacturing year, 2016–2022
miles	Numeric predictor	Odometer mileage in miles
sold_date	Context variable	Date of sale; all observations fall in August 2022
VIN / location / trim / features	Other fields	Retained in the source data but not used in the primary model

6.3 Data Quality and Variable Selection

There are no missing observations in sold price, model, year or mileage, and no duplicate rows were identified. Missing values in wheels, location, metro and state were not used in the primary model. VIN was excluded as an identification field, while country and currency provide no meaningful variation for the selected U.S. sample. Text-heavy features were also excluded from the primary regression.

6.4 Statistical Methods

The analysis used Microsoft Excel and Python. Descriptive statistics and graphs summarize the data, Pearson correlation measures linear association, simple regression examines year and mileage separately, and multiple regression estimates their combined relationship with price after controlling for model.

One-way ANOVA tests differences in mean prices across models. Because model variances are unequal, Kruskal–Wallis was also used as a non-parametric robustness check. Regression diagnostics indicated non-normality and heteroskedasticity, so results are interpreted as associations and robust standard-error checks were considered.

7. Descriptive Analysis

Variable	N	Mean	Median	SD	Min	Q1	Q3	Max
Sold Price (\$)	1,579	68,391.45	60,400	21,748.48	40,700	51,900	77,900	159,200
Mileage (miles)	1,579	26,612.87	24,228	15,808.25	1	15,424	35,192	95,803
Vehicle Year	1,579	2019.46	2019	1.21	2016	2019	2020	2022

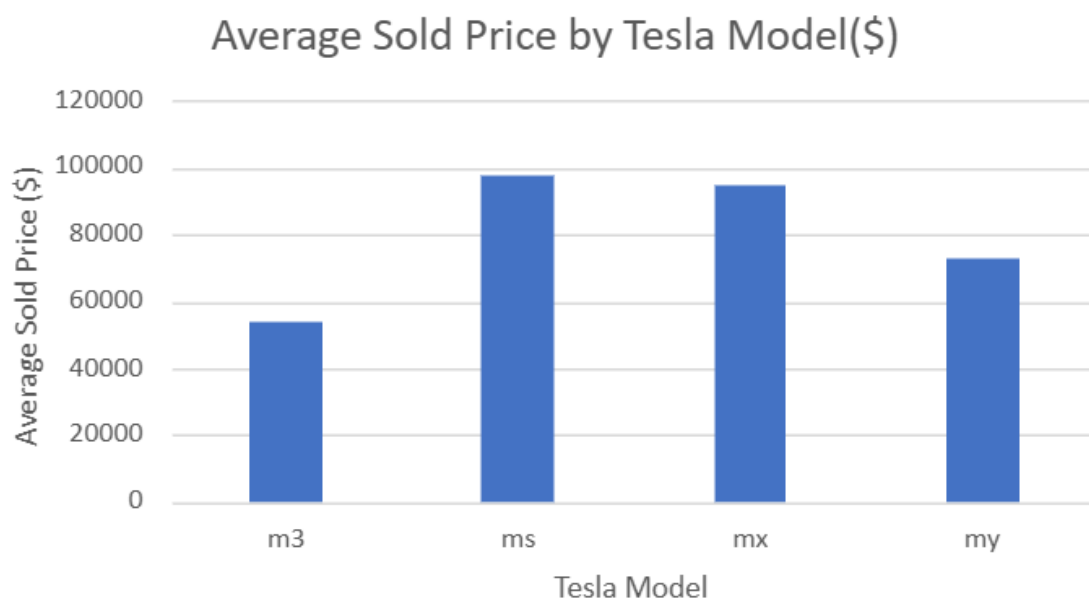
Table 2: Descriptive statistics for the core analytical variables

The average sold price is \$68,391.45, compared with a median of \$60,400. The mean being higher than the median indicates a right-skewed price distribution, with a smaller number of higher-priced vehicles pulling the mean upward. Mileage has a mean of 26,612.87 miles and a standard deviation of 15,808.25 miles, showing considerable variation in vehicle usage. Vehicle years range from 2016 to 2022, with an average of 2019.46.

Model	N	Mean Price (\$)	Median (\$)	SD (\$)
m3 / Model 3	906	53,875.50	53,000	6,279.65
ms / Model S	194	97,636.60	94,900	22,707.23
mx / Model X	239	94,863.18	92,200	17,643.05
my / Model Y	240	73,187.92	73,000	5,515.53

Table 3: Resale price by Tesla model

Figure 1: Average Sold Price by Tesla Model (user-generated Excel chart)



Year	N	Average Sold Price (\$)
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2016	9	70,922.22
2017	44	76,513.64
2018	277	63,366.79
2019	553	58,598.01
2020	339	67,399.41
2021	290	84,943.45
2022	67	97,700.00

Table 4: Average resale price by vehicle year

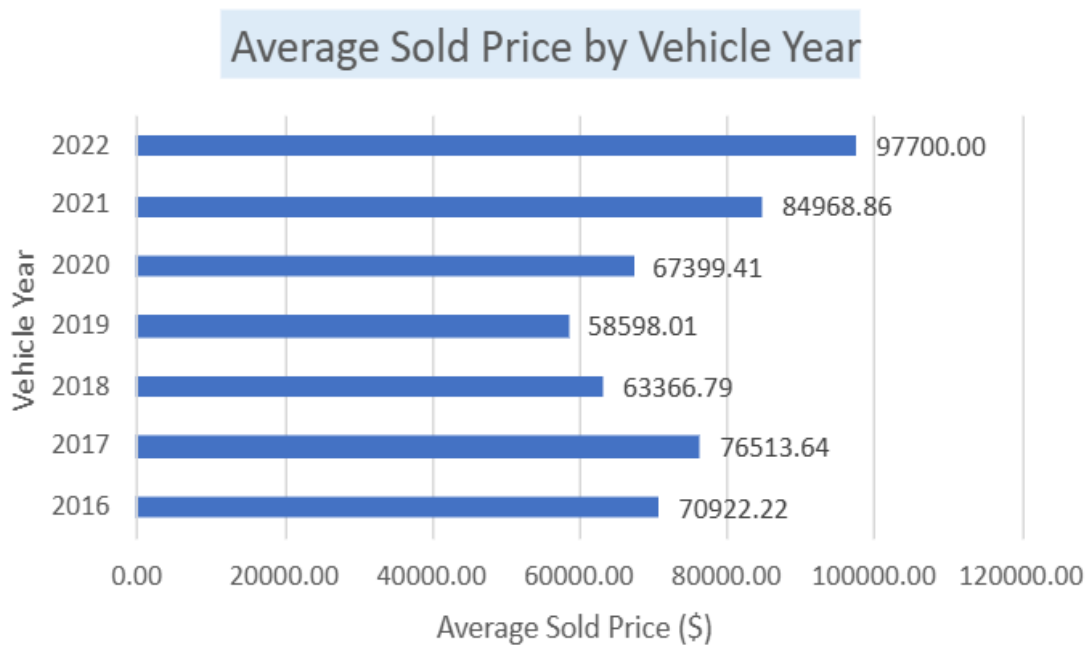


Figure 2: Average Sold Price by Vehicle Year (user-generated Excel chart)

Average price varies across manufacturing years rather than following a perfectly smooth linear pattern. The 2019 vehicles have the lowest average in the observed period at \$58,598.01, while 2022 vehicles have the highest at \$97,700.00. The higher average prices in 2021 and 2022 are consistent with the positive year-price association examined formally below.

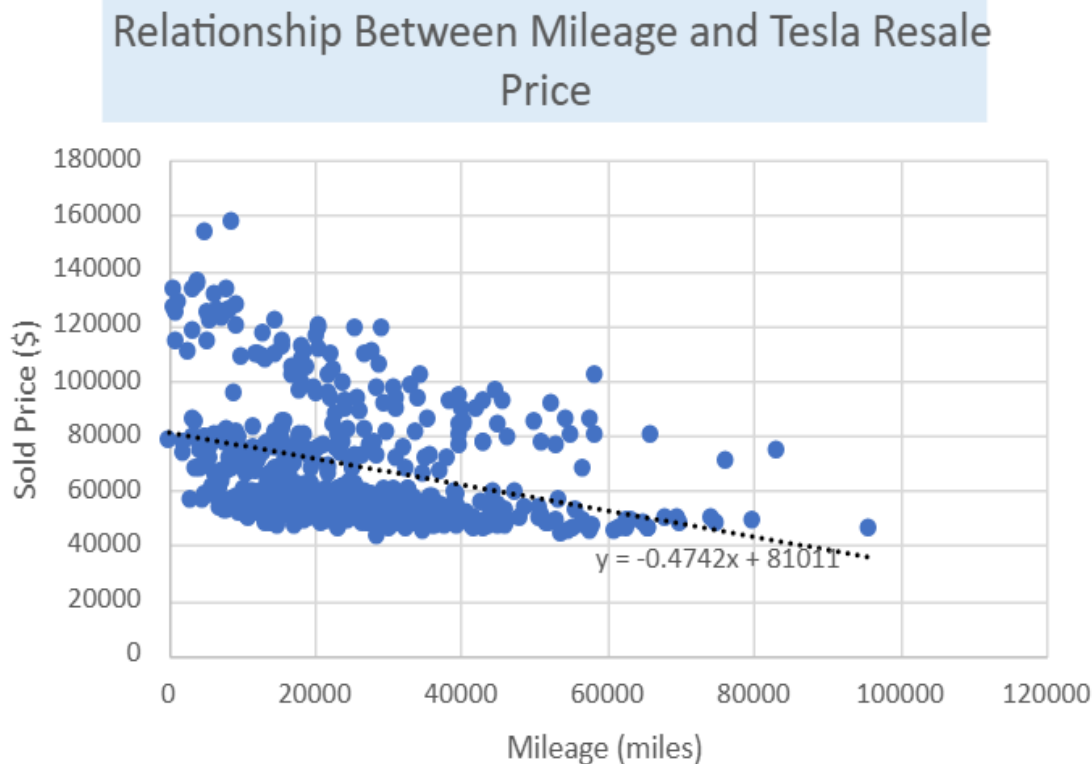


Figure 3: Relationship Between Mileage and Tesla Resale Price (user-generated Excel chart)

The mileage scatter plot shows a downward-sloping fitted line, indicating that vehicles with higher mileage tend to have lower sold prices. The scatter also shows substantial dispersion around the line, suggesting that mileage alone cannot capture all price differences. The simple regression quantifies this relationship.

8. Correlation Analysis

Variable 1	Variable 2	Pearson r	Interpretation
Year	Sold Price	0.380	Moderate positive
Mileage	Sold Price	-0.345	Moderate negative
Year	Mileage	-0.561	Moderate negative

Table 5: Pearson correlation results

Vehicle year and sold price have a moderate positive correlation ($r = 0.380$), indicating that newer vehicles in the sample tend to have higher resale prices. Mileage and sold price have a moderate negative correlation ($r = -0.345$), indicating that higher-mileage vehicles tend to sell for less. Year and mileage are moderately negatively correlated ($r = -0.561$), meaning newer vehicles in the sample generally have lower mileage. Correlation describes association and does not establish causation.

9. Simple Regression Analysis

9.1 Mileage → Sold Price

The mileage-only regression is statistically significant. Excel reports $R^2 = 0.118733$ and an F-statistic of 212.334 with Significance F = 3.23×10^{-45} . The mileage coefficient is -0.47409 ($p < 0.001$). Thus, within this simple model, an additional 1,000 miles is associated with approximately \$474 lower sold price. Mileage alone explains about 11.87% of price variation.

Figure 4: Excel Simple Regression Output — Mileage and Sold Price

SUMMARY OUTPUT									
<i>Regression Statistics</i>									
Multiple R	0.344576								
R Square	0.118733								
Adjusted R Square	0.118174								
Standard Error	20428.35								
Observations	1578								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	1	8.86E+10	8.86E+10	212.334	3.23E-45				
Residual	1576	6.58E+11	4.17E+08						
Total	1577	7.46E+11							
	Coefficient	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	81005.34	1007.22	80.42464	0	79029.71	82980.98	79029.71	82980.98	
16901	-0.47409	0.032535	-14.5717	3.23E-45	-0.5379	-0.41027	-0.5379	-0.41027	

9.2 Year → Sold Price

Simple regression	R ²	Slope coefficient	p-value
Mileage → Sold Price	0.118733	-0.47409 per mile	3.23×10^{-45}
Year → Sold Price	0.144633	+6,853.589 per year	1.81×10^{-55}

The year-only regression is also statistically significant. Excel reports $R^2 = 0.144633$, $F = 266.4844$ and Significance $F = 1.81 \times 10^{-55}$. The coefficient on year is 6,853.589 ($p < 0.001$). Therefore, a one-year increase in manufacturing year is associated with approximately \$6,854 higher sold price in the simple model. Year alone explains about 14.46% of price variation.

The two simple regressions show that both variables are individually informative, but their relatively modest R^2 values also indicate that examining year or mileage in isolation does not adequately explain the observed price differences. This motivates the multiple regression model, which additionally controls for Tesla model.

Figure 5: Excel Simple Regression Output — Vehicle Year and Sold Price

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.380307							
R Square	0.144633							
Adjusted R Square	0.14409							
Standard Error	20125.92							
Observations	1578							
ANOVA								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	1	1.08E+11	1.08E+11	266.4844	1.81E-55			
Residual	1576	6.38E+11	4.05E+08					
Total	1577	7.46E+11						
	<i>Coefficient</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	-1.4E+07	847847	-16.2437	5.59E-55	-1.5E+07	-1.2E+07	-1.5E+07	-1.2E+07
	6853.589	419.8384	16.32435	1.81E-55	6030.089	7677.09	6030.089	7677.09

10. Multiple Regression Analysis

The multiple regression model includes three model dummy variables—Model S, Model X and Model Y—with Model 3 as the reference category—along with vehicle year and mileage. The model is statistically significant ($F = 2645.176$, $p < 0.001$) and has $R^2 = 0.893708$ and adjusted $R^2 = 0.893370$. Thus, approximately 89.37% of the observed variation in sold price is explained by the selected predictors in the fitted sample.

Predictor	Coefficient (\$)	p-value	Interpretation
Model S vs Model 3	+39,888.20	<0.001	Higher price association
Model X vs Model 3	+45,138.88	<0.001	Higher price association
Model Y vs Model 3	+7,086.19	<0.001	Higher price association
Year	+7,154.94	<0.001	Positive association
Mileage	-0.18404	<0.001	Negative association

Table 6: Multiple regression coefficients

Holding year and mileage constant, Model S is associated with approximately \$39,888 higher sold price than Model 3, Model X with approximately \$45,139 higher price, and Model Y with approximately \$7,086 higher price. Holding model and mileage constant, one additional vehicle year is associated with approximately \$7,155 higher price. Holding model and year constant, each additional mile is associated with approximately \$0.184 lower price, or about \$184 for every additional 1,000 miles.

Diagnostic checks indicate that the classical regression assumptions are not fully satisfied: residual normality and constant variance are problematic in the raw OLS residuals. A robust HC3 standard-error check produced the same signs and statistical conclusions for the selected predictors. Therefore, the regression results are interpreted as strong associations in the observed sample rather than causal estimates.

Figure 6: Excel Multiple Regression Output — Model, Year and Mileage

SUMMARY OUTPUT									
Regression Statistics									
Multiple R	0.945361								
R Square	0.893708								
Adjusted R Square	0.89337								
Standard Error	7101.791								
Observations	1579								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	5	6.67E+11	1.33E+11	2645.176	0				
Residual	1573	7.93E+10	50435437						
Total	1578	7.46E+11							
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	-1.4E+07	400500.1	-35.9267	6.7E-207	-1.5E+07	-1.4E+07	-1.5E+07	-1.4E+07	
model_ms	39888.2	567.8079	70.24946	0	38774.46	41001.94	38774.46	41001.94	
model_mx	45138.88	522.9291	86.31932	0	44113.17	46164.6	44113.17	46164.6	
model_my	7086.192	572.9682	12.36751	1.35E-33	5962.331	8210.054	5962.331	8210.054	
YEAR2	7154.939	198.2394	36.09241	2.5E-208	6766.098	7543.78	6766.098	7543.78	
MILES2	-0.18404	0.01374	-13.3946	7.99E-39	-0.21099	-0.15709	-0.21099	-0.15709	

11. One-Way ANOVA

A one-way ANOVA was conducted to test whether the mean resale price differs across Model 3, Model S, Model X and Model Y. The test produced $F = 1284.473$ with $p < 0.001$. Because the p-value is below $\alpha = 0.05$, H_0 is rejected. There is statistically significant evidence that mean resale prices are not equal across all four Tesla model groups.

The descriptive means support the statistical result: Model 3 averages \$53,875.50, Model S \$97,636.60, Model X \$94,863.18 and Model Y \$73,187.92. However, ordinary ANOVA assumes reasonably similar variances. The group variances are substantially different, so a non-parametric check is appropriate rather than relying on ANOVA alone.

Figure 7: Excel One-Way ANOVA Output — Tesla Model Price Differences

Anova: Single Factor						
SUMMARY						
Groups	Count	Sum	Average	Variance		
m3	906	48811200	53875.5	39434051		
ms	194	18941500	97636.6	5.16E+08		
mx	239	22672300	94863.18	3.11E+08		
my	240	17565100	73187.92	30421067		
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	5.3E+11	3	1.77E+11	1284.473	0	2.610552
Within Groups	2.17E+11	1575	1.37E+08			
Total	7.46E+11	1578				

12. Kruskal–Wallis Non-Parametric Test

The Kruskal–Wallis test was used as a rank-based non-parametric test of whether resale-price distributions differ across the four Tesla model groups. The Python result is $H = 1161.924$ with 3 degrees of freedom and $p = 1.338 \times 10^{-251}$ ($p < 0.001$). Therefore, H_0 is rejected. The non-parametric evidence is consistent with the ANOVA finding that resale-price distributions differ across the model groups.

The Kruskal–Wallis test establishes that at least one group differs; it does not, by itself, identify every pairwise difference. Pairwise post-hoc testing was not included because the project requirement is to establish group-level differences and the multiple regression already provides model-specific comparisons against the Model 3 reference category.

Figure 8: Python Kruskal–Wallis Test Output

```

print("Kruskal-Wallis Test")
print("-----")
print(f"H statistic: {H:.3f}")
print("Degrees of freedom: 3")
print(f"p-value: {p:.3e}")

if p < 0.05:
    print("Decision: Reject H0")
else:
    print("Decision: Fail to reject H0")

```

```

Kruskal-Wallis Test
-----
H statistic: 1161.924
Degrees of freedom: 3
p-value: 1.338e-251
Decision: Reject H0

```

13. Integrated Findings

The statistical techniques tell a consistent story. Descriptive analysis shows substantial price variation across Tesla models and manufacturing years. Correlation analysis shows that newer vehicles tend to have higher prices and higher-mileage vehicles tend to have lower prices. Simple regressions confirm that both year and mileage are individually significant, although each alone explains only a modest share of price variation.

The multiple regression provides the strongest explanation because it considers model, year and mileage simultaneously. Its R^2 of 89.37% indicates that the selected characteristics jointly explain a large share of observed price variation. Model remains strongly associated with price after controlling for year and mileage.

ANOVA and Kruskal–Wallis reinforce the conclusion that resale prices differ across model groups. Overall, model, year and mileage are statistically significant factors associated with Tesla resale prices in this dataset.

Hypothesis	Test	Evidence	Decision
Mileage has no significant relationship with price	Simple regression	$p < 0.001$	Reject H_0
Year has no significant relationship with price	Simple regression	$p < 0.001$	Reject H_0
Selected factors do not explain price variation	Multiple regression	$p < 0.001$	Reject H_0
Mean prices are equal across models	ANOVA	$p < 0.001$	Reject H_0
Price distributions are the same across models	Kruskal–Wallis	$p < 0.001$	Reject H_0

Table 7: Summary of hypothesis decisions

14. Business Implications

For used-vehicle pricing, model should be treated as a primary segmentation variable rather than applying one broad price rule to all Tesla vehicles. The strong model coefficients in the multiple regression indicate that Model S, Model X and Model Y command different price levels even after accounting for year and mileage. A pricing analyst can therefore begin with model-specific reference ranges and then adjust for vehicle age and mileage.

Mileage provides a practical adjustment variable: higher mileage is associated with lower expected price. Year is also an important input because newer vehicles in the sample are associated with higher prices.

The three variables can therefore form a useful baseline pricing framework, but not a complete valuation engine because trim, battery condition, options, accident history, location and market timing are not fully represented.

15. Recommendations

R1. Build model-specific pricing bands.

Use separate benchmark ranges for Model 3, Model S, Model X and Model Y before applying year and mileage adjustments.

R2. Use year and mileage as structured adjustment variables.

Apply model-specific adjustments for vehicle year and mileage rather than a single overall depreciation rule.

R3. Develop a simple appraisal calculator.

Build a spreadsheet or dashboard combining model, year and mileage to generate a baseline expected price and flag unusual price differences for review.

R4. Add condition and specification data to future models.

Add trim, battery information, options, accident history, location and vehicle condition to future models.

R5. Refresh the model with current transactions.

Refresh the model with current transactions because the dataset is concentrated in August 2022.

16. Limitations

- Secondary, web-sourced data: the compiled dataset is obtained through Kaggle and is not presented as a direct Tesla corporate dataset.
- Time concentration: all 1,579 observations have sale dates in August 2022, limiting the ability to generalize the coefficients to other periods or current market conditions.
- Limited explanatory variables: the primary model uses model, year and mileage. Trim, options, battery condition, accident history, location and other vehicle-condition variables are not incorporated.
- Unequal model group sizes: Model 3 accounts for the largest share of observations, while Model S has a smaller sample.
- Assumption issues: regression residuals show non-normality and heteroskedasticity, and ANOVA groups have unequal variances. Robust checks and the Kruskal–Wallis test were therefore used to strengthen interpretation.
- Observational design: the results identify statistical associations and should not be interpreted as evidence that changing one characteristic would mechanically cause the observed price change.

17. Conclusion

This study evaluated used Tesla resale prices using 1,579 vehicle-level observations. The analysis indicates that model, manufacturing year and mileage are statistically significant factors associated with sold price. Multiple regression explains approximately 89.37% of observed price variation, while ANOVA and Kruskal–Wallis indicate significant differences across models.

The findings support a model-specific pricing framework in which year and mileage are used as structured adjustments. A more complete valuation system should also incorporate vehicle condition, trim, battery-related information, location and current market conditions.

18. References

- Saturn Data Cloud. Daily Used Tesla Car Sales for United States. Dataset accessed through Kaggle.
- Kaggle. Daily Used Tesla Car Sales for United States. <https://www.kaggle.com/datasets/saturndatacloud/daily-used-tesla-car-sales-for-united-states>
- Microsoft. Use the Analysis ToolPak to perform complex data analysis in Excel.
- SciPy documentation. `scipy.stats.kruskal` — Kruskal–Wallis H-test for independent samples.

Annexure A: Data Description

The raw dataset contains 1,579 observations and 16 columns: vin, year, model, color, miles, trim, sold_price, interior, wheels, features, country, location, metro, state, currency and sold_date. The raw file is retained separately as the source dataset used for the analysis.

Data-quality item	Result	Action / interpretation
Core variables missing	0	No imputation required for model, year, miles or sold price
Wheels missing	34	Excluded from primary model
Location missing	145	Excluded from primary model
Metro missing	239	Excluded from primary model
State missing	15	Excluded from primary model
Duplicate rows	0	No duplicate rows identified
Sale dates	31 unique dates	All within August 2022

Table A1: Data-quality checks

Annexure B: Sample Questionnaire

Because the project uses secondary data, a sample questionnaire is included to satisfy the project brief. The questionnaire is designed to collect primary perceptions relevant to used-Tesla pricing and the factors considered by buyers when evaluating resale value. It is not used to generate the statistical results in this report.

- 1. Which Tesla model would you consider purchasing in the used-vehicle market? (Model 3 / Model S / Model X / Model Y)
- 2. What vehicle age would you consider acceptable for a used Tesla? (0–2 years / 3–4 years / 5–6 years / More than 6 years)
- 3. What mileage range would you consider acceptable? (Below 20,000 / 20,000–40,000 / 40,001–60,000 / Above 60,000 miles)
- 4. How important is mileage when evaluating a used Tesla's price? (1 = Not important, 5 = Extremely important)
- 5. How important is Tesla model when evaluating resale price? (1 = Not important, 5 = Extremely important)
- 6. How important is vehicle manufacturing year when evaluating resale price? (1 = Not important, 5 = Extremely important)
- 7. Which factor would most influence the price you are willing to pay? (Model / Year / Mileage / Condition / Features / Location)
- 8. Would you pay more for a newer Tesla with higher mileage than an older Tesla with lower mileage? (Yes / No / Depends on condition)
- 9. How useful would an online Tesla resale-price calculator be when buying a used Tesla? (1 = Not useful, 5 = Very useful)
- 10. What additional information should be included in a used-Tesla pricing model? (Open-ended)